

Amrutha Vinayakam

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SUMMARY

Data Scientist and AI/ML Engineer with 5+ years of experience in AI/ML engineering, predictive modeling, and data platform development. Built LLM solutions to operationalize policy-driven eligibility and workflow automation. Improved monthly revenue projections by 20% by building financial forecasting models from historical claims and billing data for chronic care programs. Spearheaded 5+ ETL pipelines, cutting instructional preparation time by 40%.

PROFESSIONAL EXPERIENCE

Data Science Manager | Food Bank of South Jersey | Pennsauken, NJ

Dec 2025 – Present

- Architected a **centralized BigQuery data warehouse**, reducing data discrepancies by 15% and **saving 8 hours/week** by designing scalable schemas, ingestion processes, and standardized reporting layers.
- Built an ensemble of **Random Forest and Logistic Regression models** to segment agencies based on demographic and behavioral data, driving a **15% increase in distribution**.
- Delivered **executive-ready Power BI dashboards and KPI reporting**, enabling weekly and monthly operational reviews by translating program needs into governed metrics and reusable analytics views.
- Fostered **Cross-Functional Collaboration (Grants, Marketing, Finance)** by aligning stakeholders on KPIs, translating data insights into actionable strategies, and supporting data-driven decision-making across programs.
- Deployed an **XGBoost model for demand forecasting**, leveraging historical distribution and program data to predict trends, improving planning accuracy and enabling better resource allocation across agencies.

Artificial Intelligence Engineer | Dreamline AI | Florida, Remote

Jul 2025 – Dec 2025

- Improved user behavior prediction by 25% by developing an **ensemble of XGBoost and Logistic Regression** models using user, property, and policy data, with **LLM-based feature enrichment** for program rules, while optimizing for latency and cost.
- Increased match accuracy by 10% by segmenting users using **K-Means clustering** for program alignment, validated using t-tests to ensure statistical significance of key insights.
- Enhanced **production chatbot's intelligence by integrating OpenAI API**, achieving a 15% boost in response accuracy.
- Engineered an **automated evaluation suite using F1, confusion matrices, and cross-validation**, ensuring model reliability across releases; maintained a 95% testing coverage rate.

Graduate Teaching/Technical Assistant | Fordham University GSAS | NY

Sept 2023 – Dec 2024

- Developed and maintained 5+ **ETL data pipelines** for academic datasets, reducing data quality issues by 40% by standardizing ingestion, cleaning, and data-validation steps for 30+ students.
- Created 4+ **interactive Tableau dashboards** for feedback and enrollment KPIs, improving faculty decision-making efficiency by 50% by consolidating metrics into self-serve visual reporting.
- Implemented a **Transformers-based classification model**, achieving 92% accuracy by fine-tuning Hugging Face models to automate research content tagging and save 10+ hours/month.

Artificial Intelligence Engineer | Radical AI | New York, NY

May 2024 – Oct 2024

- Enhanced model quality by 25% and real-time processing speed by 30% by iterating on feature pipelines and training workflows using Python, SQL, Scikit-learn, TensorFlow, PyTorch, and Keras.
- Boosted production chatbot response accuracy by 15% by enhancing NLP pipelines with **Hugging Face Transformers and OpenAI API**, including prompt and retrieval tuning.
- Reduced inference latency by 20% by preprocessing and optimizing text pipelines using SpaCy and NLTK for large-scale datasets.
- Optimized pipeline efficiency by 35% by integrating **LangChain and LlamaIndex** components to increase throughput and support scalable agent workflows.
- Validated model reliability to 98% accuracy by implementing evaluation suites (F1, confusion matrices, cross-validation) and tracking regressions across releases.
- Drove up user support engagement by 15% by contributing to AI virtual agent features that expanded real-time assistance coverage.

Data Analyst | KPMG | India

Aug 2020 – Jul 2023

- Improved monthly revenue projections by 20% by building financial forecasting models from historical claims and billing data for chronic care programs.
- Improved model deployment by deploying forecasting models (**ARIMAX, SARIMAX**) and XGBoost in Azure ML Studio for repeatable training and scoring, and automated monthly runs to support revenue planning.
- Developed and presented ROI models to cross-functional finance stakeholders, influencing **\$2M+ investment decisions** across 3 digital health pilots.
- Reduced billing delays by 15% by developing SQL-based reports that monitored revenue-cycle performance across 7 departments.
- Reduced annual overhead by \$250K by automating Excel dashboards that tracked cost per patient by service line and surfaced inefficiencies.
- Achieved 8% cost savings by designing A/B tests to compare pricing and contract scenarios, improving confidence in pharmaceutical negotiation strategies.

TECHNICAL SKILLS

Programming: Python (scikit-learn, pytorch, pyspark, numpy, pandas, matplotlib, causalimpact), SQL

Analytics: Machine Learning, A/B Testing, Causal Inference, Predictive Modeling, Regression, Classification, Clustering

Tools: Snowflake, MongoDB, Google BigQuery, Azure, GCP, AWS, Databricks, Power BI, Tableau, Looker, MLOps

EDUCATION

Fordham University, New York — Master of Science in Data Science

Aug 2023 – May 2025